

**Top 15 Important Physics Questions for B.E. Students
(2024-2025 Batch) by Dr. Md. Raheem Ahmed |
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1. Explain construction and working of different types of lasers (CO₂ and Semiconductor)
2. Explain construction and working of Laser Induced Break down spectroscopy.
3. Explain how optical fiber is manufactured using double crucible method
4. Explain the construction and working of Pressure sensor
5. Explain the working Magnetostriction method
6. What is Hall effect? Derive the expression for hall coefficient.
7. Explain the construction and working of Solar Cell.
8. Derive the Maxwell integral form to differential form.
9. What is Domain Theory? Derive the expression for Molecular field theory.
10. What is Meissner effect? Explain the concept of BCS theory.
11. What is de-Broglie wave length? Derive the expression Schrodinger time independent wave equation.
12. Derive the expression for energy of the particle in one dimensional potential box and write conclusions of Kroning- Penney model.
13. What are Quantum Bits? Explain the CNOT theorem.
14. Explain different types of fabrications of nano materials (Ball Milling and Sol-Gel method.
15. Explain the construction and working of Scanning Electron Microscopy.

Important top 15 Short questions

- 1. What is Stimulated Emission? Explain some applications of LASER.**
- 2. What is basic principle of optical fibre(TIR)?**
- 3. Give short note on different types of optical fiber based on refractive index profile and give its applications.**
- 4. Give some engineering applications of Ultrasonic waves.**
- 5. What are QLED's?**
- 6. What is Piezoelectric Generator?**
- 7. Write any four difference of Type-I and Type-II superconductors.**
- 8. What are High Tc-Superconductors?**
- 9. Explain Physical significance wave function.**
- 10. Explain short note on Hadamard Gate.**
- 11. Give some applications of Quantum computing.**
- 12. What is the use of surface to volume ratio at nano scale.**
- 13. What is the difference between bulk and thin film.**
- 14. Write short note on thermal evaporation method in thin films.**
- 15. Write applications of thin film in engineering.**